

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Tx V_SWING limits adjustment
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

Change Gen4 V_SWING range to 410 – 545 mV peak (note that the material presented in the Cairo Electrical WG defined 410 – 540 mV peak and the high limit was slightly fine-tuned since then)
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Benefits as a result of the changes:

The change is needed for obtaining alignment between the compliance test setup and the functional link setup, for the following reasons:
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| <ol style="list-style-type: none">1) The measurement is performed using an oscilloscope terminated with 50Ω, which is higher than typical Rx termination that is closer to 42.5Ω2) USB Type-C systems may contain additional serial resistance on the high-speed paths due to EOS reasons (typically up to 4Ω single-ended). |
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An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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None

An analysis of the hardware implications:
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None

An analysis of the software implications:
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None

An analysis of the compliance testing implications:
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None

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Actual Change

(a). Section 3.2.3 (Table 3-22)

From Text:

Table 3-22. Transmitter Specifications for Gen 4 (at TP2)

Symbol	Description	Min	Max	Units	Comments
UI	Minimum Unit Interval	39.0508	39.0742	ps	Corresponding to the Gen 4 baseline Baud rate of 25.6GB with an uncertainty range of -300 ppm to 300 ppm. See Notes 1, 2.
V_SWING	Peak differential voltage swing	390	500	mV	See Note 9 and Section 3.2.3.2.
TX_LEVELS_MISMATCH	Levels separation mismatch ratio	0.975	--	--	See Note 9 and Section 3.2.3.3.
TX_SNR	Signal to Noise and Distortion Ratio	32.5	--	dB	See Note 9 and Section 3.2.3.4.

To Text:

Table 3-22. Transmitter Specifications for Gen 4 (at TP2)

Symbol	Description	Min	Max	Units	Comments
UI	Minimum Unit Interval	39.0508	39.0742	ps	Corresponding to the Gen 4 baseline Baud rate of 25.6GB with an uncertainty range of -300 ppm to 300 ppm. See Notes 1, 2.
V_SWING	Peak differential voltage swing	390410	500545	mV	See Note 9 and Section 3.2.3.2.
TX_LEVELS_MISMATCH	Levels separation mismatch ratio	0.975	--	--	See Note 9 and Section 3.2.3.3.
TX_SNR	Signal to Noise and Distortion Ratio	32.5	--	dB	See Note 9 and Section 3.2.3.4.

(b). Section 3.2.4.2 (Table 3-27)

From Text:

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Table 3-27. Stressed Signal for Gen 4 Receiver Compliance Testing

Test Case	Voltage Swing [mV pk-pk]	SNDR [dB]	Level Mismatch	ACCM Noise [mV Pk-Pk]	PJ [UI Pk-Pk]	RJ [UI RMS]
1	1000	32.5	0.975	100	0.085	0.0085
2a+2b	800	32.5	0.975	100	0.075	0.0085

To Text:

Table 3-27. Stressed Signal for Gen 4 Receiver Compliance Testing

Test Case	Voltage Swing [mV pk-pk]	SNDR [dB]	Level Mismatch	ACCM Noise [mV Pk-Pk]	PJ [UI Pk-Pk]	RJ [UI RMS]
1	1000 1090	32.5	0.975	100	0.085	0.0085
2a+2b	800 820	32.5	0.975	100	0.075	0.0085

(c). Section 3.2.5.2 (Table 3-28)

From Text:

Table 3-28. Captive Device Transmitter Specifications for Gen 4 (at TP3)

Symbol	Description	Min	Max	Units	Comments
UI	Minimum Unit Interval	39.0508	39.0742	ps	Corresponding to the Gen 4 baseline Baud rate of 25.6GB with an uncertainty range of -300 ppm to 300 ppm. See Notes 1, 2.
V_SWING	Peak differential voltage swing	350	500	mV	See Notes 9, 10
TX_LEVELS_MISMATCH	Levels separation mismatch ratio	0.975	--	--	See Note 9 and Section 3.2.3.3.
TX_SNR	Signal to Noise and Distortion Ratio	32.5	--	dB	See Note 9 and Section 3.2.3.4.

To Text:

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Table 3-28. Captive Device Transmitter Specifications for Gen 4 (at TP3)

Symbol	Description	Min	Max	Units	Comments
UI	Minimum Unit Interval	39.0508	39.0742	ps	Corresponding to the Gen 4 baseline Baud rate of 25.6GB with an uncertainty range of -300 ppm to 300 ppm. See Notes 1, 2.
V_SWING	Peak differential voltage swing	350 410	500 545	mV	See Notes 9, 10
TX_LEVELS_MISMATCH	Levels separation mismatch ratio	0.975	--	--	See Note 9 and Section 3.2.3.3.
TX_SNDR	Signal to Noise and Distortion Ratio	32.5	--	dB	See Note 9 and Section 3.2.3.4.

(d). Section 3.2.5.3.2 (Table 3-30)

From Text:

Table 3-30. Stressed Signal for Gen 4 Captive Device Receiver Compliance Testing

Voltage Swing [mV pk-pk]	SNDR [dB]	Level Mismatch	ACCM Noise [mV Pk-Pk]	PJ [UI Pk-Pk]	RJ [UI RMS]
800	32.5	0.975	100	0.075	0.0085

To Text:

Table 3-30. Stressed Signal for Gen 4 Captive Device Receiver Compliance Testing

Voltage Swing [mV pk-pk]	SNDR [dB]	Level Mismatch	ACCM Noise [mV Pk-Pk]	PJ [UI Pk-Pk]	RJ [UI RMS]
800 820	32.5	0.975	100	0.075	0.0085